AGV-SAFE

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Automated Guided Vehicle (AGV) Safety Controller

Industrial Mobile Robotics / Functional Safety | Rev A2 - Documentation Baseline 21 Jul 2026 | Architecture, Atomic Requirements, Interfaces & Verification

Mission profile. The controller provides collision-avoidance and motion-safety functions for warehouse AGVs through dual-channel supervision of LiDAR, encoder and safety-I/O information. It enforces Safe Torque Off (STO) and Safe Limited Speed (SLS) according to the approved machine safety concept, while navigation mission planning and ordinary traction control remain external. SIL 3 / PL e is treated as a release objective whose scope must be proven by the responsible program.

Engineering baseline. The architecture uses 1oo2D dual-channel logic, a safety-certified RTOS, 48 V mobile power, LiDAR/encoder processing, and safety communication through the released PROFINET Safety or Fail-Safe over EtherCAT (FSoE) variant. Safety response is verified from fault onset through sensing, computation, cross-monitoring, communication, STO/SLS output and actual vehicle stopping behavior. Manufacturing includes rigorous safety-I/O fault injection, while the 15-year lifecycle preserves safety-stop event records for hazard analytics without permitting analytics to auto-modify safety parameters.

Vehicle mass, braking system, safety scanner, encoder, drive interface, communication profile, RTOS/MCU, safety parameter or certified component changes trigger explicit safety impact assessment and only the necessary but sufficient revalidation; unresolved evidence remains a release blocker.

Definition	Controlled product interpretation
Product class	Dedicated motion-safety controller for warehouse AGVs and autonomous material-handling vehicles.
Safety / assurance	Baseline safety intent: dual-channel 1oo2D architecture with SIL 3 / PL e STO and SLS release objective, supported by project-specific certification evidence and end-to-end stopping verification.
Documentation baseline	21 Jul 2026 - controlled PRD baseline date requested for this product.
Primary architecture	Dual-channel safety controller, safety-certified RTOS, LiDAR and encoder inputs, STO/SLS outputs, 48 V DC input, and PROFINET Safety or FSoE communication variants.
Release philosophy	Safety timing, independence, machine interfaces, fault injection, certification scope and manufacturing genealogy must be objectively closed for each released AGV variant. Quantitative safety parameters including SLS speed envelope, STO FTTI, stopping distance, AGV mass/payload, braking/deceleration envelope, scanner timing, protective-field margin and proof/diagnostic intervals remain controlled program parameters and block final release until frozen and evidenced.

PRODUCT GOVERNANCE, SAFETY CASE & REQUIREMENT QUALITY

The AGV safety controller is governed as a motion-safety product whose safety functions, diagnostics, communication profiles and machine interfaces are configuration controlled. Safety claims are tied to hazard analysis, safe-state definition, validated timing and objective certification evidence.

A safety label or certified component cannot substitute for a complete system safety argument. Any change to stopping dynamics, sensor architecture, drive interface or safety communication can invalidate assumptions that appear unrelated in software.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-001	The AGV safety controller is governed as a motion-safety product whose safety functions, diagnostics, communication profiles and machine interfaces are configuration controlled.	Safety plan, hazard/risk traceability, controlled parameter register, certification evidence map and release matrix.
AGVSAFE-002	Release shall remain blocked if the safety argument relies only on a safety label or component certificate without system-level hazard, timing, integration and assumption-of-use evidence.	Safety plan, hazard/risk traceability, controlled parameter register, certification evidence map and release matrix.
AGVSAFE-003	Changes to safety function, AGV dynamics, sensor, drive interface, network profile or certification scope require documented impact analysis and closure of the affected regression evidence before release.	Safety plan, hazard/risk traceability, controlled parameter register, certification evidence map and release matrix.

Verification, margin and future-change rule

Changes affecting safety function, AGV dynamics, sensor, drive interface, network profile or certification scope require documented impact analysis against the mechanism on this page; reusable evidence is retained only within its demonstrated similarity boundary.

SYSTEM BOUNDARY, AGV / DRIVE / BRAKE OWNERSHIP & EXCLUSIONS

The controller owns safety sensing, safe-speed evaluation, STO/SLS command generation and assigned safety communication. Traction torque control, mechanical brake sizing, navigation mission planning and warehouse traffic orchestration remain external unless specifically allocated.

Boundary ambiguity can create dual authority or a gap between a safety demand and actual torque removal. The ICD therefore identifies electrical, logical and timing ownership for every safety-relevant path.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-004	The controller owns safety sensing, safe-speed evaluation, STO/SLS command generation and assigned safety communication. This behavior shall be preserved for every released variant to which this section applies.	System boundary diagram, safety interface matrix, drive/brake ICD and authority review.
AGVSAFE-005	When the following failure condition occurs - Boundary ambiguity can create dual authority or a gap between a safety demand and actual torque removal - the system shall preserve diagnosability and prevent unqualified normal operation.	System boundary diagram, safety interface matrix, drive/brake ICD and authority review.
AGVSAFE-006	The verification record for this mechanism shall contain System boundary diagram, safety interface matrix, drive/brake ICD and authority review.	System boundary diagram, safety interface matrix, drive/brake ICD and authority review.

Verification, margin and future-change rule

Reopen this verification topic whenever drive inverter, brake controller, navigation computer, safety scanner or vehicle architecture changes. The responsible engineer identifies moved assumptions and closes the resulting analysis or DVP before approval.

HAZARD SCENARIOS, COLLISION-AVOIDANCE INTENT & SAFE STATE

The released hazard set covers collision, overspeed, unintended motion, loss of perception, incorrect direction, loss of braking/torque-removal authority and unsafe restart according to the AGV application. Safe state and degraded states are defined per hazard and operating context.

Stopping is not always the only safe response; for example, loss of a noncritical sensor may permit bounded low-speed recovery if the safety concept explicitly allows it. The controller never invents degraded authority from convenience logic.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-007	The released hazard set covers collision, overspeed, unintended motion, loss of perception, incorrect direction, loss of braking/torque-removal authority and unsafe restart according to the AGV application.	Hazard analysis, safety goals, safe-state table and scenario-based review.
AGVSAFE-008	Fault analysis shall include Stopping is not always the only safe response; for example, loss of a noncritical sensor may permit bounded low-speed recovery if the safety concept explicitly allows it. Continued normal authority requires that the relevant validity checks remain satisfied.	Hazard analysis, safety goals, safe-state table and scenario-based review.
AGVSAFE-009	Changes involving warehouse layout, vehicle mass/speed, braking capability, hazard zone or safety goal shall be assessed against Hazard analysis, safety goals, safe-state table and scenario-based review before the configuration is accepted.	Hazard analysis, safety goals, safe-state table and scenario-based review.

Verification, margin and future-change rule

The change-control trigger here is any modification to warehouse layout, vehicle mass/speed, braking capability, hazard zone or safety goal. Acceptance requires restored margin, interface consistency and objective evidence for the affected mechanism.

OPERATING MODES, STARTUP, SERVICE & RESTART AUTHORITY

Modes include powered-off, boot/self-test, safe standstill, automatic travel, manual/service movement, degraded operation, fault-latched state and authorized reset. Motion permission is issued only after required channels, communication and output-path diagnostics are valid.

Unsafe restart after a transient fault is a primary concern. Reset authority therefore requires cause removal, valid safety inputs and any operator acknowledgement defined by the machine safety concept.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-010	Across all applicable operating states, Modes include powered-off, boot/self-test, safe standstill, automatic travel, manual/service movement, degraded operation, fault-latched state and authorized reset.	State-machine test, startup fault injection, service-mode review and restart interlock verification.
AGVSAFE-011	After a transient safety fault, motion re-enable shall remain inhibited until the released reset and restart conditions are satisfied.	State-machine test, startup fault injection, service-mode review and restart interlock verification.
AGVSAFE-012	State transitions shall prevent motion enable until startup, service-mode and restart interlocks are simultaneously valid.	State-machine test, startup fault injection, service-mode review and restart interlock verification.

Verification, margin and future-change rule

For future variants, a change in mode logic, reset policy, service controls, maintenance procedure or motion-enable interface is an engineering event, not an editorial update; the impact record defines evidence reuse and targeted regression.

48 V DC INPUT, PROTECTION & MOBILE POWER INTERFACE

The controller operates from the AGV 48 V battery system through a protected input stage whose normal, transient, reverse-polarity and brownout limits are program-controlled. Safety output behavior remains deterministic during supply decay.

A brownout can cause asynchronous channel resets or unintended output pulses. Power-good, reset and output-enable sequencing are therefore included in the safety timing analysis.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-013	The controller operates from the AGV 48 V battery system through a protected input stage whose normal, transient, reverse-polarity and brownout limits are program-controlled.	Programmable supply sweep, reverse/transient test, brownout timing capture and restart verification.
AGVSAFE-014	The controller shall recognize or bound A brownout can cause asynchronous channel resets or unintended output pulses. Response priority shall follow the mechanism-specific safety or data-validity rule.	Programmable supply sweep, reverse/transient test, brownout timing capture and restart verification.
AGVSAFE-015	Targeted regression for this topic shall use Programmable supply sweep, reverse/transient test, brownout timing capture and restart verification whenever battery system, DC/DC stage, input protector, connector or supply transient requirement changes.	Programmable supply sweep, reverse/transient test, brownout timing capture and restart verification.

Verification, margin and future-change rule

Changes to battery system, DC/DC stage, input protector, connector or supply transient requirement are screened against the failure mechanism described above, with only demonstrably unaffected evidence carried forward.

BROWNOUT, HOLD-UP & SAFE OUTPUT DECAY

Loss of supply is detected early enough to place safety outputs in the defined de-energized or fail-safe state before logic becomes uncontrolled. Hold-up, if provided, is used for deterministic shutdown and diagnostics rather than prolonging unsafe actuation.

Different channel undervoltage thresholds can create disagreement or delayed STO. Cross-channel monitoring includes power-domain health where it affects the safety function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-016	Loss of supply is detected early enough to place safety outputs in the defined de-energized or fail-safe state before logic becomes uncontrolled.	Brownout ramp test, dual-channel threshold comparison and safe-output decay measurement.
AGVSAFE-017	The design shall detect and contain: Different channel undervoltage thresholds can create disagreement or delayed STO. The product shall make the loss of validity observable.	Brownout ramp test, dual-channel threshold comparison and safe-output decay measurement.
AGVSAFE-018	The engineering evidence set shall include Brownout ramp test, dual-channel threshold comparison and safe-output decay measurement and identify the exact released variant.	Brownout ramp test, dual-channel threshold comparison and safe-output decay measurement.

Verification, margin and future-change rule

If PMIC, reset device, hold-up capacitance, output driver or undervoltage threshold changes, the release owner documents the affected assumptions, selects the required regression and updates the evidence index before sign-off.

DUAL-CHANNEL SAFETY ARCHITECTURE & INDEPENDENCE

Two safety channels independently acquire assigned inputs, execute safety logic and supervise the output path according to the released architecture. Shared resources are identified and justified so a single fault cannot silently defeat both channels.

Common clock, power, connector, memory or software-service dependencies can defeat apparent redundancy. Independence analysis therefore covers physical and systematic common causes, not only duplicate processors.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-019	Two safety channels independently acquire assigned inputs, execute safety logic and supervise the output path according to the released architecture.	Architecture FMEA/FMEDA as applicable, independence review and single-fault injection.
AGVSAFE-020	The design shall prevent silent acceptance of Common clock, power, connector, memory or software-service dependencies can defeat apparent redundancy.	Architecture FMEA/FMEDA as applicable, independence review and single-fault injection.
AGVSAFE-021	Revalidation after changes to processor, power rail, clock, connector, memory or shared software component shall close Architecture FMEA/FMEDA as applicable, independence review and single-fault injection.	Architecture FMEA/FMEDA as applicable, independence review and single-fault injection.

Verification, margin and future-change rule

Variant evolution involving processor, power rail, clock, connector, memory or shared software component requires a traceable change-impact review that identifies what remains valid and what must be reverified.

1oo2D CROSS-MONITORING, DISAGREEMENT & DIAGNOSTIC COVERAGE

The 1oo2D architecture cross-compares channel results and diagnostic state with a bounded reaction to disagreement. Voting semantics, diagnostic test intervals and the conditions that permit continued operation are explicit.

A latent fault in one channel followed by a second fault is addressed through periodic diagnostics and fault latching consistent with the claimed integrity target.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-022	The 1oo2D architecture cross-compares channel results and diagnostic state with a bounded reaction to disagreement.	Channel disagreement test, latent-fault injection and diagnostic-interval verification.
AGVSAFE-023	If this mechanism occurs - A latent fault in one channel followed by a second fault is addressed through periodic diagnostics and fault latching consistent with the claimed integrity target - the product shall retain a deterministic and diagnosable response.	Channel disagreement test, latent-fault injection and diagnostic-interval verification.
AGVSAFE-024	The function shall not be considered verified without Channel disagreement test, latent-fault injection and diagnostic-interval verification.	Channel disagreement test, latent-fault injection and diagnostic-interval verification.

Verification, margin and future-change rule

The verification baseline is reopened when cross-monitoring logic, comparator, diagnostic interval, voter or channel implementation changes enough to alter the stated behavior, margin, interface or failure response.

SAFETY MCU / LOGIC, CLOCKS, RESET & MEMORY PROTECTION

Safety compute uses supervised clocks, reset sources and memory-integrity mechanisms appropriate to the safety allocation. Startup self-test establishes a known state before safety outputs can enable motion.

Clock drift, memory corruption or reset asymmetry can produce plausible but divergent calculations. Hardware diagnostics and software plausibility detect these conditions before authority is retained.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-025	Across applicable modes, Safety compute uses supervised clocks, reset sources and memory-integrity mechanisms appropriate to the safety allocation.	Clock fault injection, memory/ECC test, reset-sequence test and startup self-test evidence.
AGVSAFE-026	Fault handling shall cover Clock drift, memory corruption or reset asymmetry can produce plausible but divergent calculations. Normal authority shall not persist by assumption alone.	Clock fault injection, memory/ECC test, reset-sequence test and startup self-test evidence.
AGVSAFE-027	Evidence reuse after a change to MCU, memory, clock source, reset supervisor or compiler/runtime is permitted only where Clock fault injection, memory/ECC test, reset-sequence test and startup self-test evidence remains representative.	Clock fault injection, memory/ECC test, reset-sequence test and startup self-test evidence.

Verification, margin and future-change rule

Any modification of MCU, memory, clock source, reset supervisor or compiler/runtime is compared with the released assumptions on this page; unproven similarity results in targeted re-analysis or retest.

SAFETY INPUT CONDITIONING, DIGITAL I/O & PLAUSIBILITY

Safety digital inputs include controlled thresholds, filtering, diagnostics and wiring-fault detection appropriate to their function. Input debounce never exceeds the allocated reaction-time budget.

Open circuit, short to supply/ground, welded contact or chattering input must be distinguishable where required by the safety concept.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-028	Safety digital inputs include controlled thresholds, filtering, diagnostics and wiring-fault detection appropriate to their function.	Electrical fault injection, threshold sweep, debounce timing and wiring-diagnostic test.
AGVSAFE-029	The design shall enter the allocated controlled response when Open circuit, short to supply/ground, welded contact or chattering input must be distinguishable where required by the safety concept.	Electrical fault injection, threshold sweep, debounce timing and wiring-diagnostic test.
AGVSAFE-030	The release record shall demonstrate closure against the function-specific verification evidence and controlled acceptance criteria.	Electrical fault injection, threshold sweep, debounce timing and wiring-diagnostic test.

Verification, margin and future-change rule

Future changes in input circuit, connector, filter, sensor type or diagnostic method are accepted only after the configuration record explains evidence reuse, new risks and the regression needed for closure.

STO OUTPUT PATH, DRIVER DIAGNOSTICS & TORQUE-REMOVAL INTERFACE

STO outputs use independent or appropriately monitored channels to remove drive torque through the allocated inverter interface. Output diagnostics cover stuck-on, stuck-off and cross-short conditions to the extent required by the safety concept.

A logical STO command is not sufficient evidence of torque removal; interface timing and drive acknowledgement, if credited, are validated end-to-end.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-031	After startup and during normal operation, STO outputs use independent or appropriately monitored channels to remove drive torque through the allocated inverter interface.	STO electrical test, stuck-output injection, drive-interface timing and torque-removal evidence.
AGVSAFE-032	The design shall explicitly handle A logical STO command is not sufficient evidence of torque removal; interface timing and drive acknowledgement, if credited, are validated end-to-end. The condition shall be diagnosed or bounded before normal authority is retained.	STO electrical test, stuck-output injection, drive-interface timing and torque-removal evidence.
AGVSAFE-033	A change to output driver, drive inverter, STO interface, wiring or feedback method shall trigger impact analysis and targeted regression using STO electrical test, stuck-output injection, drive-interface timing and torque-removal evidence.	STO electrical test, stuck-output injection, drive-interface timing and torque-removal evidence.

Verification, margin and future-change rule

Engineering change control for output driver, drive inverter, STO interface, wiring or feedback method preserves the intent of this page by linking the modified design to updated analysis, test evidence and variant identity.

ENCODER INPUTS, DIRECTION & SPEED PLAUSIBILITY

Encoder channels provide speed and direction information with controlled scaling, edge timing and plausibility checks. Redundant or diverse speed sources are compared according to the released architecture.

Missing pulses, extra pulses, reversed polarity or frozen counts can understate actual speed. The SLS function reacts based on validated data rather than trusting a single numerical value.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-034	Encoder channels provide speed and direction information with controlled scaling, edge timing and plausibility checks.	Encoder simulator tests, missing/extra pulse injection, direction reversal and speed-correlation test.
AGVSAFE-035	The product shall not accept as valid normal operation: Missing pulses, extra pulses, reversed polarity or frozen counts can understate actual speed. It shall enter the section-defined invalid, restricted or protective response.	Encoder simulator tests, missing/extra pulse injection, direction reversal and speed-correlation test.
AGVSAFE-036	Objective release evidence for the stated limits and assumptions shall include Encoder simulator tests, missing/extra pulse injection, direction reversal and speed-correlation test.	Encoder simulator tests, missing/extra pulse injection, direction reversal and speed-correlation test.

Verification, margin and future-change rule

A change to encoder type, resolution, interface circuit, gear ratio or wheel geometry shall not inherit prior approval automatically; the affected evidence is reviewed for continued representativeness before release.

LiDAR INTERFACE, SAFETY ZONES & DATA VALIDITY

LiDAR safety data is accepted only from released sensor/interface variants with controlled scan/status semantics and freshness. Zone definitions, field geometry and protective distance are parameters owned by the application safety design.

Loss of scan, stale point data, blinded optics or invalid status cannot be interpreted as a clear path. The controller transitions to the allocated safe/degraded response.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-037	LiDAR safety data is accepted only from released sensor/interface variants with controlled scan/status semantics and freshness.	Sensor disconnect/stale test, zone intrusion test, blinded-sensor case and field-geometry verification.
AGVSAFE-038	Loss of scan, stale point data, blinded optics or invalid status cannot be interpreted as a clear path. This condition shall trigger the diagnostic or containment behavior allocated by this section instead of silent continuation.	Sensor disconnect/stale test, zone intrusion test, blinded-sensor case and field-geometry verification.
AGVSAFE-039	Changes to LiDAR model, field configuration, mounting position, network/interface or protective zone require documented impact analysis and closure of the affected regression evidence before release.	Sensor disconnect/stale test, zone intrusion test, blinded-sensor case and field-geometry verification.

Verification, margin and future-change rule

Changes affecting LiDAR model, field configuration, mounting position, network/interface or protective zone require documented impact analysis against the mechanism on this page; reusable evidence is retained only within its demonstrated similarity boundary.

POINT-CLOUD / RANGE PREPROCESSING & SAFETY DATA REDUCTION

If raw LiDAR data is processed by the controller, the reduction from scan data to safety-relevant occupancy or distance information is deterministic and traceable. Filtering never removes an obstacle solely to improve navigation convenience.

Numerical overflow, dropped packets or excessive filtering can shrink the apparent obstacle set. Diagnostics identify incomplete frames and invalid preprocessing state.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-040	If raw LiDAR data is processed by the controller, the reduction from scan data to safety-relevant occupancy or distance information is deterministic and traceable. This behavior shall be preserved for every released variant to which this section applies.	Golden-scene replay, packet-loss injection, boundary-object test and algorithm traceability.
AGVSAFE-041	When the following failure condition occurs - Numerical overflow, dropped packets or excessive filtering can shrink the apparent obstacle set - the system shall preserve diagnosability and prevent unqualified normal operation.	Golden-scene replay, packet-loss injection, boundary-object test and algorithm traceability.
AGVSAFE-042	The verification record for this mechanism shall contain Golden-scene replay, packet-loss injection, boundary-object test and algorithm traceability.	Golden-scene replay, packet-loss injection, boundary-object test and algorithm traceability.

Verification, margin and future-change rule

Reopen this verification topic whenever LiDAR data format, filter, point-cloud library, frame rate or preprocessing algorithm changes. The responsible engineer identifies moved assumptions and closes the resulting analysis or DVP before approval.

SAFE SPEED ESTIMATION & VEHICLE-KINEMATIC INPUTS

Safe speed is calculated from validated encoder or other allocated motion inputs using controlled wheel diameter, gearing and direction parameters. The safety function distinguishes measured speed from navigation-commanded speed.

Calibration error or wheel slip can create a gap between commanded and actual motion. The safety case identifies any credited diversity or margin used to address this.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-043	Safe speed is calculated from validated encoder or other allocated motion inputs using controlled wheel diameter, gearing and direction parameters.	Speed-rig test, parameter variation, slip scenario analysis and independent speed correlation.
AGVSAFE-044	Fault analysis shall include Calibration error or wheel slip can create a gap between commanded and actual motion. Continued normal authority requires that the relevant validity checks remain satisfied.	Speed-rig test, parameter variation, slip scenario analysis and independent speed correlation.
AGVSAFE-045	Changes involving wheel/tire, gearing, encoder scaling, motion sensor or calibration shall be assessed against Speed-rig test, parameter variation, slip scenario analysis and independent speed correlation before the configuration is accepted.	Speed-rig test, parameter variation, slip scenario analysis and independent speed correlation.

Verification, margin and future-change rule

The change-control trigger here is any modification to wheel/tire, gearing, encoder scaling, motion sensor or calibration. Acceptance requires restored margin, interface consistency and objective evidence for the affected mechanism.

SAFE LIMITED SPEED (SLS) FUNCTION & LIMIT ENFORCEMENT

SLS monitors actual motion against the application-defined safe-speed limit and initiates the allocated response when the limit is exceeded or reliable speed cannot be established. The baseline SIL 3 / PL e claim is treated as a release objective requiring project evidence.

Threshold hysteresis, measurement delay and braking/torque-removal time are included in the full SLS response budget; a limit value alone is not a complete safety requirement.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-046	Across all applicable operating states, SLS monitors actual motion against the application-defined safe-speed limit and initiates the allocated response when the limit is exceeded or reliable speed cannot be established.	SLS threshold/ramp test, overspeed injection and end-to-end response-time measurement.
AGVSAFE-047	The SLS response budget shall include threshold hysteresis, measurement delay and braking/torque-removal time; the speed limit alone is not a complete safety requirement.	SLS threshold/ramp test, overspeed injection and end-to-end response-time measurement.
AGVSAFE-048	SLS timing margin shall be demonstrated over threshold-ramp and overspeed conditions using end-to-end response-time measurement.	SLS threshold/ramp test, overspeed injection and end-to-end response-time measurement.

Verification, margin and future-change rule

For future variants, a change in SLS limit, sensing path, filtering, response strategy or certification allocation is an engineering event, not an editorial update; the impact record defines evidence reuse and targeted regression.

SAFE TORQUE OFF (STO) FUNCTION & DEMAND RESPONSE

STO demand removes propulsion torque through the released hardware path within the allocated fault-tolerant response time. Software supervision may request STO, but the path does not depend solely on high-level navigation or non-safety communications.

Failure to remove torque, delayed gate inhibit or re-energization without reset are verified as distinct failure modes.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-049	STO demand removes propulsion torque through the released hardware path within the allocated fault-tolerant response time.	STO demand test, response-time capture, restart interlock test and inverter interface verification.
AGVSAFE-050	The controller shall recognize or bound Failure to remove torque, delayed gate inhibit or re-energization without reset are verified as distinct failure modes. Response priority shall follow the mechanism-specific safety or data-validity rule.	STO demand test, response-time capture, restart interlock test and inverter interface verification.
AGVSAFE-051	Targeted regression for this topic shall use STO demand test, response-time capture, restart interlock test and inverter interface verification whenever STO path, inverter, wiring, reset semantics or power stage changes.	STO demand test, response-time capture, restart interlock test and inverter interface verification.

Verification, margin and future-change rule

Changes to STO path, inverter, wiring, reset semantics or power stage are screened against the failure mechanism described above, with only demonstrably unaffected evidence carried forward.

BRAKING, COAST-DOWN & DRIVE INTERACTION

The safety controller coordinates its demands with the drive and mechanical/electrical braking architecture defined by the AGV integrator. It does not assume a stopping distance from STO alone unless coast-down and brake behavior justify that claim.

A high-friction floor, low-friction floor, payload change or brake degradation can materially alter stopping distance. These external dependencies are controlled in the machine safety case.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-052	The safety controller coordinates its demands with the drive and mechanical/electrical braking architecture defined by the AGV integrator.	Dynamic stopping tests across approved mass/speed/surface conditions and brake-state fault cases.
AGVSAFE-053	The design shall detect and contain: A high-friction floor, low-friction floor, payload change or brake degradation can materially alter stopping distance. The product shall make the loss of validity observable.	Dynamic stopping tests across approved mass/speed/surface conditions and brake-state fault cases.
AGVSAFE-054	The engineering evidence set shall include Dynamic stopping tests across approved mass/speed/surface conditions and brake-state fault cases and identify the exact released variant.	Dynamic stopping tests across approved mass/speed/surface conditions and brake-state fault cases.

Verification, margin and future-change rule

If vehicle mass, tire, floor, brake system, regenerative braking or drive torque response changes, the release owner documents the affected assumptions, selects the required regression and updates the evidence index before sign-off.

STOPPING-DISTANCE MODEL, PROTECTIVE FIELD & MARGIN

Protective field sizing accounts for vehicle speed, perception/communication delay, safety-controller reaction, drive/brake response and physical stopping distance with the program-defined margin.

A field that is adequate at nominal payload may be unsafe at maximum mass or degraded braking. Parameter sets are therefore tied to released vehicle variants.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-055	Protective field sizing accounts for vehicle speed, perception/communication delay, safety-controller reaction, drive/brake response and physical stopping distance with the program-defined margin.	Worst-case stop-distance calculation, measured correlation and zone-boundary validation.
AGVSAFE-056	The design shall prevent silent acceptance of A field that is adequate at nominal payload may be unsafe at maximum mass or degraded braking.	Worst-case stop-distance calculation, measured correlation and zone-boundary validation.
AGVSAFE-057	Revalidation after changes to vehicle variant, payload, speed limit, brake response, sensor delay or field margin shall close Worst-case stop-distance calculation, measured correlation and zone-boundary validation.	Worst-case stop-distance calculation, measured correlation and zone-boundary validation.

Verification, margin and future-change rule

Variant evolution involving vehicle variant, payload, speed limit, brake response, sensor delay or field margin requires a traceable change-impact review that identifies what remains valid and what must be reverified.

OBSTACLE ZONES, DIRECTIONAL FIELDS & MODE DEPENDENCE

Safety zones may vary with direction, speed and approved operating mode only through validated parameter sets. Field switching is deterministic and does not create a transient unprotected region.

Incorrect direction status or late field switching can move the protective boundary behind the actual hazard.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-058	Safety zones may vary with direction, speed and approved operating mode only through validated parameter sets.	Direction-change test, field-switch timing capture and boundary-intrusion scenarios.
AGVSAFE-059	If this mechanism occurs - Incorrect direction status or late field switching can move the protective boundary behind the actual hazard - the product shall retain a deterministic and diagnosable response.	Direction-change test, field-switch timing capture and boundary-intrusion scenarios.
AGVSAFE-060	The function shall not be considered verified without Direction-change test, field-switch timing capture and boundary-intrusion scenarios.	Direction-change test, field-switch timing capture and boundary-intrusion scenarios.

Verification, margin and future-change rule

The verification baseline is reopened when zone set, direction input, speed band, mode or LiDAR configuration changes enough to alter the stated behavior, margin, interface or failure response.

SENSOR FUSION / PLAUSIBILITY BETWEEN LiDAR AND ENCODERS

Where multiple sensing channels contribute to motion safety, plausibility rules define what disagreement means and which information can retain authority. Navigation estimates are not automatically treated as safety-grade evidence.

A plausible but stale navigation velocity must not override a safety encoder indicating overspeed. Freshness and source integrity are explicit.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-061	Across applicable modes, Where multiple sensing channels contribute to motion safety, plausibility rules define what disagreement means and which information can retain authority.	Cross-sensor disagreement injection, stale-data tests and authority-table review.
AGVSAFE-062	Fault handling shall cover A plausible but stale navigation velocity must not override a safety encoder indicating overspeed. Normal authority shall not persist by assumption alone.	Cross-sensor disagreement injection, stale-data tests and authority-table review.
AGVSAFE-063	Evidence reuse after a change to sensor set, fusion logic, freshness limit, navigation interface or source priority is permitted only where Cross-sensor disagreement injection, stale-data tests and authority-table review remains representative.	Cross-sensor disagreement injection, stale-data tests and authority-table review.

Verification, margin and future-change rule

Any modification of sensor set, fusion logic, freshness limit, navigation interface or source priority is compared with the released assumptions on this page; unproven similarity results in targeted re-analysis or retest.

LOSS OF LiDAR / OBSCURATION / COMMUNICATION FAULT

Loss, obstruction or invalid status of required LiDAR data triggers the safe/degraded response allocated by the hazard analysis. Automatic recovery requires restored valid scans and any required reset conditions.

Repeated intermittent obscuration can create oscillation between safe and operational states; recovery timing and latching prevent unsafe chatter.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-064	Loss, obstruction or invalid status of required LiDAR data triggers the safe/degraded response allocated by the hazard analysis.	Obscuration test, disconnect/reconnect, intermittent packet loss and recovery-state verification.
AGVSAFE-065	The design shall enter the allocated controlled response when Repeated intermittent obscuration can create oscillation between safe and operational states; recovery timing and latching prevent unsafe chatter.	Obscuration test, disconnect/reconnect, intermittent packet loss and recovery-state verification.
AGVSAFE-066	The release record shall demonstrate closure against the function-specific verification evidence and controlled acceptance criteria.	Obscuration test, disconnect/reconnect, intermittent packet loss and recovery-state verification.

Verification, margin and future-change rule

Future changes in LiDAR, cleaning concept, network link, recovery policy or fault latch are accepted only after the configuration record explains evidence reuse, new risks and the regression needed for closure.

LOSS OF ENCODER / SPEED-SOURCE DISAGREEMENT

If required speed data is missing or inconsistent, the controller cannot simply assume zero motion. SLS and related functions transition according to the approved fallback or STO strategy.

Single-channel plausibility that masks a frozen encoder is specifically challenged in fault injection.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-067	After startup and during normal operation, If required speed data is missing or inconsistent, the controller cannot simply assume zero motion.	Frozen/stuck encoder tests, channel disagreement, missing pulses and safe-response timing.
AGVSAFE-068	The design shall explicitly handle Single-channel plausibility that masks a frozen encoder is specifically challenged in fault injection. The condition shall be diagnosed or bounded before normal authority is retained.	Frozen/stuck encoder tests, channel disagreement, missing pulses and safe-response timing.
AGVSAFE-069	A change to encoder, redundant speed source, plausibility thresholds or fallback strategy shall trigger impact analysis and targeted regression using Frozen/stuck encoder tests, channel disagreement, missing pulses and safe-response timing.	Frozen/stuck encoder tests, channel disagreement, missing pulses and safe-response timing.

Verification, margin and future-change rule

Engineering change control for encoder, redundant speed source, plausibility thresholds or fallback strategy preserves the intent of this page by linking the modified design to updated analysis, test evidence and variant identity.

COMMON-CAUSE FAULT ANALYSIS & DIVERSITY

Common-cause analysis addresses shared power, environment, connector damage, software tools, calibration, sensor mounting and communication dependencies. Diversity is credited only where implementation and failure modes support it.

Two channels built from identical assumptions can fail together under a systematic error even when electrically separate.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-070	Common-cause analysis addresses shared power, environment, connector damage, software tools, calibration, sensor mounting and communication dependencies.	CCF analysis, design review, environmental correlation and selected dual-fault scenarios.
AGVSAFE-071	The product shall not accept as valid normal operation: Two channels built from identical assumptions can fail together under a systematic error even when electrically separate. It shall enter the section-defined invalid, restricted or protective response.	CCF analysis, design review, environmental correlation and selected dual-fault scenarios.
AGVSAFE-072	Objective release evidence for the stated limits and assumptions shall include CCF analysis, design review, environmental correlation and selected dual-fault scenarios.	CCF analysis, design review, environmental correlation and selected dual-fault scenarios.

Verification, margin and future-change rule

A change to shared component, design tool, calibration dataset, mounting or environmental assumption shall not inherit prior approval automatically; the affected evidence is reviewed for continued representativeness before release.

DIAGNOSTIC COVERAGE, LATENT FAULTS & PROOF TESTS

Diagnostics target safety-relevant latent faults in input, compute, communication and output paths with intervals compatible with the safety claim. Any required proof test or maintenance activity is documented for the operator.

A diagnostic that is disabled during normal operation or never exercised in the field cannot be credited without evidence.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-073	Diagnostics target safety-relevant latent faults in input, compute, communication and output paths with intervals compatible with the safety claim.	Diagnostic coverage analysis, periodic-test timing and maintenance/proof-test validation.
AGVSAFE-074	A diagnostic that is disabled during normal operation or never exercised in the field cannot be credited without evidence. This condition shall trigger the diagnostic or containment behavior allocated by this section instead of silent continuation.	Diagnostic coverage analysis, periodic-test timing and maintenance/proof-test validation.
AGVSAFE-075	Changes to diagnostic interval, built-in test, service procedure or credited coverage require documented impact analysis and closure of the affected regression evidence before release.	Diagnostic coverage analysis, periodic-test timing and maintenance/proof-test validation.

Verification, margin and future-change rule

Changes affecting diagnostic interval, built-in test, service procedure or credited coverage require documented impact analysis against the mechanism on this page; reusable evidence is retained only within its demonstrated similarity boundary.

WATCHDOGS, TASK SUPERVISION & SAFE RECOVERY

Independent watchdogs supervise safety tasks, communication health and cross-channel progress. Recovery does not automatically restore motion; the safe-state and reset policy remain authoritative.

A watchdog that resets both channels simultaneously can become a common-cause mechanism. Reset architecture and retained diagnostics are reviewed accordingly.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-076	Independent watchdogs supervise safety tasks, communication health and cross-channel progress. This behavior shall be preserved for every released variant to which this section applies.	Task-stall injection, watchdog timeout test, reset-cause retention and restart interlock verification.
AGVSAFE-077	When the following failure condition occurs - A watchdog that resets both channels simultaneously can become a common-cause mechanism - the system shall preserve diagnosability and prevent unqualified normal operation.	Task-stall injection, watchdog timeout test, reset-cause retention and restart interlock verification.
AGVSAFE-078	The verification record for this mechanism shall contain Task-stall injection, watchdog timeout test, reset-cause retention and restart interlock verification.	Task-stall injection, watchdog timeout test, reset-cause retention and restart interlock verification.

Verification, margin and future-change rule

Reopen this verification topic whenever watchdog source, timeout, reset topology or task architecture changes. The responsible engineer identifies moved assumptions and closes the resulting analysis or DVP before approval.

SAFETY RTOS, TASK PARTITIONING & SCHEDULING

The safety-certified RTOS configuration allocates priority, memory, timing and communication resources so safety tasks meet worst-case deadlines under released load. Non-safety diagnostics cannot starve safety execution.

Priority inversion, unbounded blocking or dynamic allocation can erode reaction-time margin. Such mechanisms are either bounded or excluded from safety paths.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-079	The safety-certified RTOS configuration allocates priority, memory, timing and communication resources so safety tasks meet worst-case deadlines under released load.	WCET/scheduling analysis, stress load, priority-inversion test and RTOS configuration audit.
AGVSAFE-080	Fault analysis shall include Priority inversion, unbounded blocking or dynamic allocation can erode reaction-time margin. Continued normal authority requires that the relevant validity checks remain satisfied.	WCET/scheduling analysis, stress load, priority-inversion test and RTOS configuration audit.
AGVSAFE-081	Changes involving RTOS version, scheduler, task set, IPC mechanism or memory policy shall be assessed against WCET/scheduling analysis, stress load, priority-inversion test and RTOS configuration audit before the configuration is accepted.	WCET/scheduling analysis, stress load, priority-inversion test and RTOS configuration audit.

Verification, margin and future-change rule

The change-control trigger here is any modification to RTOS version, scheduler, task set, IPC mechanism or memory policy. Acceptance requires restored margin, interface consistency and objective evidence for the affected mechanism.

WCET, END-TO-END FTTI & TIMING BUDGET

Safety timing budgets decompose sensing, filtering, computation, cross-channel comparison, communication, output actuation and drive/brake response. FTTI acceptance is tied to the hazard-specific safety concept.

Average task time is not used as proof of safe reaction. Worst-case jitter and diagnostic delays are included.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-082	Across all applicable operating states, Safety timing budgets decompose sensing, filtering, computation, cross-channel comparison, communication, output actuation and drive/brake response.	Timing instrumentation, WCET evidence and end-to-end fault-to-safe-state measurement.
AGVSAFE-083	Safety timing evidence shall use WCET and end-to-end fault-response bounds; average task time shall not be used as proof of safe reaction.	Timing instrumentation, WCET evidence and end-to-end fault-to-safe-state measurement.
AGVSAFE-084	The released schedule shall meet WCET and fault-to-safe-state timing budgets under worst-case concurrent load.	Timing instrumentation, WCET evidence and end-to-end fault-to-safe-state measurement.

Verification, margin and future-change rule

For future variants, a change in processor, software load, sensor rate, network timing, output driver or FTTI allocation is an engineering event, not an editorial update; the impact record defines evidence reuse and targeted regression.

FREEDOM FROM INTERFERENCE & MIXED-CRITICALITY SERVICES

Safety functions are protected from non-safety logging, analytics, service interfaces and maintenance communication through memory, CPU, I/O and network resource controls appropriate to the architecture.

A non-safety buffer overflow or diagnostic flood must not delay STO/SLS execution.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-085	Safety functions are protected from non-safety logging, analytics, service interfaces and maintenance communication through memory, CPU, I/O and network resource controls appropriate to the architecture.	Resource-exhaustion test, memory-protection test and mixed-criticality interference analysis.
AGVSAFE-086	The controller shall recognize or bound A non-safety buffer overflow or diagnostic flood must not delay STO/SLS execution. Response priority shall follow the mechanism-specific safety or data-validity rule.	Resource-exhaustion test, memory-protection test and mixed-criticality interference analysis.
AGVSAFE-087	Targeted regression for this topic shall use Resource-exhaustion test, memory-protection test and mixed-criticality interference analysis whenever diagnostic service, logging rate, memory map, CPU budget or service interface changes.	Resource-exhaustion test, memory-protection test and mixed-criticality interference analysis.

Verification, margin and future-change rule

Changes to diagnostic service, logging rate, memory map, CPU budget or service interface are screened against the failure mechanism described above, with only demonstrably unaffected evidence carried forward.

PROFINET SAFETY PROFILE, TELEGRAM VALIDITY & WATCHDOG

Where the released variant uses the baseline PROFINET Safety option, safety communication implements the selected profile with controlled connection identity, sequence/freshness and watchdog timing. Exact protocol profile and certification scope remain program-controlled.

Lost, delayed, repeated or misrouted safety data causes the connection-specific safe response rather than acceptance of stale values.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-088	Where the released variant uses the baseline PROFINET Safety option, safety communication implements the selected profile with controlled connection identity, sequence/freshness and watchdog timing.	Protocol conformance test, delay/drop/replay injection and watchdog timing verification.
AGVSAFE-089	The design shall detect and contain: Lost, delayed, repeated or misrouted safety data causes the connection-specific safe response rather than acceptance of stale values. The product shall make the loss of validity observable.	Protocol conformance test, delay/drop/replay injection and watchdog timing verification.
AGVSAFE-090	The engineering evidence set shall include Protocol conformance test, delay/drop/replay injection and watchdog timing verification and identify the exact released variant.	Protocol conformance test, delay/drop/replay injection and watchdog timing verification.

Verification, margin and future-change rule

If network stack, safety profile, watchdog, device identity or topology changes, the release owner documents the affected assumptions, selects the required regression and updates the evidence index before sign-off.

FSoE PROFILE, CONNECTION ID & SAFE COMMUNICATION

For FSoE variants, connection identifiers, sequence handling, CRC/safety-code semantics and timeout behavior are configuration controlled and isolated from ordinary EtherCAT traffic assumptions.

Incorrect routing or repeated telegrams must not preserve safety authority beyond the allowed communication interval.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-091	For FSoE variants, connection identifiers, sequence handling, CRC/safety-code semantics and timeout behavior are configuration controlled and isolated from ordinary EtherCAT traffic assumptions.	FSoE conformance test, fault injection, topology change test and timeout capture.
AGVSAFE-092	The design shall prevent silent acceptance of Incorrect routing or repeated telegrams must not preserve safety authority beyond the allowed communication interval.	FSoE conformance test, fault injection, topology change test and timeout capture.
AGVSAFE-093	Revalidation after changes to FSoE stack, EtherCAT master/slave, connection ID, topology or cycle time shall close FSoE conformance test, fault injection, topology change test and timeout capture.	FSoE conformance test, fault injection, topology change test and timeout capture.

Verification, margin and future-change rule

Variant evolution involving FSoE stack, EtherCAT master/slave, connection ID, topology or cycle time requires a traceable change-impact review that identifies what remains valid and what must be reverified.

SAFETY COMMUNICATION SELECTION & VARIANT CONTROL

PROFINET Safety and FSoE are treated as separate released communication variants, not interchangeable runtime options. Each variant has its own timing, diagnostics, certification and integration evidence.

A field substitution from one safety network to another without complete requalification is prohibited.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-094	PROFINET Safety and FSoE are treated as separate released communication variants, not interchangeable runtime options.	Variant release matrix, communication-specific DVP and configuration audit.
AGVSAFE-095	The affected hardware or software variant shall remain unreleased while required qualification or production evidence for this topic is open.	Variant release matrix, communication-specific DVP and configuration audit.
AGVSAFE-096	Production release is contingent on completion of Variant release matrix, communication-specific DVP and configuration audit.	Variant release matrix, communication-specific DVP and configuration audit.

Verification, margin and future-change rule

The verification baseline is reopened when selected safety network, firmware, network controller or safety communication library changes enough to alter the stated behavior, margin, interface or failure response.

CYBERSECURITY BOUNDARY & SAFETY INDEPENDENCE

Cybersecurity controls protect configuration, service access and network integrity without delaying local safety mechanisms. Safety functions fail safely under loss of nonessential IT/OT services.

A compromised maintenance account must not be able to alter protected safety parameters or suppress STO/SLS diagnostics outside the authorized safety-change process.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-097	Across applicable modes, Cybersecurity controls protect configuration, service access and network integrity without delaying local safety mechanisms.	Access-control test, parameter tamper attempt and safety-under-network-loss verification.
AGVSAFE-098	Fault handling shall cover A compromised maintenance account must not be able to alter protected safety parameters or suppress STO/SLS diagnostics outside the authorized safety-change process. Normal authority shall not persist by assumption alone.	Access-control test, parameter tamper attempt and safety-under-network-loss verification.
AGVSAFE-099	Evidence reuse after a change to service account, remote access, network segmentation, credential policy or update mechanism is permitted only where Access-control test, parameter tamper attempt and safety-under-network-loss verification remains representative.	Access-control test, parameter tamper attempt and safety-under-network-loss verification.

Verification, margin and future-change rule

Any modification of service account, remote access, network segmentation, credential policy or update mechanism is compared with the released assumptions on this page; unproven similarity results in targeted re-analysis or retest.

SECURE / AUTHORIZED SAFETY PARAMETER MANAGEMENT

Safety parameters such as speed limits, field sets, scaling and timing values are versioned, range checked and changed only through authorized procedures with traceability.

A syntactically valid but wrong vehicle variant parameter can be hazardous; configuration identity is checked against the machine build.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-100	Safety parameters such as speed limits, field sets, scaling and timing values are versioned, range checked and changed only through authorized procedures with traceability.	Parameter range test, wrong-variant negative test, authorization audit and restore procedure.
AGVSAFE-101	The design shall enter the allocated controlled response when A syntactically valid but wrong vehicle variant parameter can be hazardous; configuration identity is checked against the machine build.	Parameter range test, wrong-variant negative test, authorization audit and restore procedure.
AGVSAFE-102	The release record shall demonstrate closure against the function-specific verification evidence and controlled acceptance criteria.	Parameter range test, wrong-variant negative test, authorization audit and restore procedure.

Verification, margin and future-change rule

Future changes in safety parameter, machine variant, commissioning tool or authorization process are accepted only after the configuration record explains evidence reuse, new risks and the regression needed for closure.

SERVICE MODE, MANUAL MOVEMENT & MAINTENANCE BYPASS CONTROL

Service or manual movement modes constrain speed and safety-function behavior according to the approved maintenance concept. Any bypass is deliberate, time/condition bounded and visibly indicated where required.

A maintenance bypass that persists after power cycle or mode exit is treated as a safety defect.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-103	After startup and during normal operation, Service or manual movement modes constrain speed and safety-function behavior according to the approved maintenance concept.	Service-mode test, bypass persistence test and operator-indication review.
AGVSAFE-104	The design shall explicitly handle A maintenance bypass that persists after power cycle or mode exit is treated as a safety defect. The condition shall be diagnosed or bounded before normal authority is retained.	Service-mode test, bypass persistence test and operator-indication review.
AGVSAFE-105	A change to service interface, bypass mechanism, mode selector or maintenance procedure shall trigger impact analysis and targeted regression using Service-mode test, bypass persistence test and operator-indication review.	Service-mode test, bypass persistence test and operator-indication review.

Verification, margin and future-change rule

Engineering change control for service interface, bypass mechanism, mode selector or maintenance procedure preserves the intent of this page by linking the modified design to updated analysis, test evidence and variant identity.

EVENT LOGGING, SAFETY-STOP RECORDS & TIME INTEGRITY

The baseline requires continuous logging of safety-stop events for warehouse hazard analytics. Records preserve event type, active safety state, relevant sensor/communication diagnostics, configuration identity and timestamp while respecting storage/privacy policy.

Event logging does not participate in the real-time safety decision and cannot delay the safe response. Loss of logging is diagnosed separately.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-106	The baseline requires continuous logging of safety-stop events for warehouse hazard analytics.	Event generation/replay test, power-loss retention and log-throughput stress test.
AGVSAFE-107	The product shall not accept as valid normal operation: Event logging does not participate in the real-time safety decision and cannot delay the safe response. It shall enter the section-defined invalid, restricted or protective response.	Event generation/replay test, power-loss retention and log-throughput stress test.
AGVSAFE-108	Objective release evidence for the stated limits and assumptions shall include Event generation/replay test, power-loss retention and log-throughput stress test.	Event generation/replay test, power-loss retention and log-throughput stress test.

Verification, margin and future-change rule

A change to log schema, storage, time source, retention policy or analytics backend shall not inherit prior approval automatically; the affected evidence is reviewed for continued representativeness before release.

NVM, CONFIGURATION INTEGRITY & POWER-LOSS RECOVERY

Safety configuration and event records use integrity controls and atomic update mechanisms so interrupted writes cannot leave a plausible but unintended parameter set.

On integrity failure, the controller rejects the affected configuration and remains in the approved restricted state until restored.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-109	Safety configuration and event records use integrity controls and atomic update mechanisms so interrupted writes cannot leave a plausible but unintended parameter set.	Power-cut write test, CRC/integrity corruption, backup/restore and invalid-config startup test.
AGVSAFE-110	On integrity failure, the controller rejects the affected configuration and remains in the approved restricted state until restored. This condition shall trigger the diagnostic or containment behavior allocated by this section instead of silent continuation.	Power-cut write test, CRC/integrity corruption, backup/restore and invalid-config startup test.
AGVSAFE-111	Changes to NVM technology, file format, parameter store or update mechanism require documented impact analysis and closure of the affected regression evidence before release.	Power-cut write test, CRC/integrity corruption, backup/restore and invalid-config startup test.

Verification, margin and future-change rule

Changes affecting NVM technology, file format, parameter store or update mechanism require documented impact analysis against the mechanism on this page; reusable evidence is retained only within its demonstrated similarity boundary.

MECHANICAL PACKAGING, SHOCK / VIBRATION & MOBILE ROBOT MOUNTING

The enclosure, PCB retention, connectors and mounting withstand the high shock/vibration environment assigned to mobile robotics by the program. Exact profiles are controlled by the AGV installation and qualification plan.

Mechanical damage that intermittently opens a safety channel or changes sensor/reference grounding is included in post-stress functional verification.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-112	The enclosure, PCB retention, connectors and mounting withstand the high shock/vibration environment assigned to mobile robotics by the program. This condition is mandatory for the affected release.	Shock/vibration qualification and post-stress full safety-I/O retest.
AGVSAFE-113	Release status for this topic shall remain open until all required objective evidence is complete and approved for the exact configuration.	Shock/vibration qualification and post-stress full safety-I/O retest.
AGVSAFE-114	This topic remains open in the release matrix until Shock/vibration qualification and post-stress full safety-I/O retest is approved.	Shock/vibration qualification and post-stress full safety-I/O retest.

Verification, margin and future-change rule

Reopen this verification topic whenever enclosure, mounting bracket, connector, PCB stack or vehicle vibration profile changes. The responsible engineer identifies moved assumptions and closes the resulting analysis or DVP before approval.

THERMAL MANAGEMENT, COMPONENT DERATING & HOT RESTART

Thermal design maintains safety compute, input/output drivers and communication components within released ratings across the AGV duty cycle. Thermal monitoring does not depend on navigation software.

Overtemperature response preserves safety outputs and avoids a reset pattern that could accidentally re-enable motion.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-115	Thermal design maintains safety compute, input/output drivers and communication components within released ratings across the AGV duty cycle.	Thermal chamber test, hot restart, component temperature correlation and fault-response verification.
AGVSAFE-116	Fault analysis shall include Overtemperature response preserves safety outputs and avoids a reset pattern that could accidentally re-enable motion. Continued normal authority requires that the relevant validity checks remain satisfied.	Thermal chamber test, hot restart, component temperature correlation and fault-response verification.
AGVSAFE-117	Changes involving enclosure, power dissipation, component, thermal interface or ambient profile shall be assessed against Thermal chamber test, hot restart, component temperature correlation and fault-response verification before the configuration is accepted.	Thermal chamber test, hot restart, component temperature correlation and fault-response verification.

Verification, margin and future-change rule

The change-control trigger here is any modification to enclosure, power dissipation, component, thermal interface or ambient profile. Acceptance requires restored margin, interface consistency and objective evidence for the affected mechanism.

EMC / ESD, MOTOR-NOISE IMMUNITY & SIGNAL ROBUSTNESS

EMC design addresses traction inverter, motor, charger and warehouse RF disturbances at safety inputs, sensors, power and network interfaces. The qualification objective includes correct safety response during disturbance, not merely survival.

A transient that corrupts one channel while leaving another valid is handled through diagnostics and cross-monitoring.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-118	Across all applicable operating states, EMC design addresses traction inverter, motor, charger and warehouse RF disturbances at safety inputs, sensors, power and network interfaces.	Conducted/radiated immunity, ESD, EFT/surge as applicable and live safety-function monitoring.
AGVSAFE-119	A transient that corrupts one channel while another remains valid shall be detected and contained through channel diagnostics and cross-monitoring.	Conducted/radiated immunity, ESD, EFT/surge as applicable and live safety-function monitoring.
AGVSAFE-120	EMC and transient immunity shall preserve channel diagnostics and the allocated safe reaction without undetected loss of safety authority.	Conducted/radiated immunity, ESD, EFT/surge as applicable and live safety-function monitoring.

Verification, margin and future-change rule

For future variants, a change in harness, filter, connector, shielding, grounding or drive system EMC environment is an engineering event, not an editorial update; the impact record defines evidence reuse and targeted regression.

CONNECTORS, HARNESS, GROUNDING & SAFETY CHANNEL SEPARATION

Connector pinout, harness routing, grounding and channel separation preserve independence and diagnostic assumptions. Shared return paths and shield connections are reviewed for common-cause effects.

Miswiring or connector partial engagement is detected where required rather than silently changing safety input interpretation.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-121	Connector pinout, harness routing, grounding and channel separation preserve independence and diagnostic assumptions.	Harness/connector ICD review, miswire test, retention test and channel isolation measurement.
AGVSAFE-122	The controller shall recognize or bound Miswiring or connector partial engagement is detected where required rather than silently changing safety input interpretation. Response priority shall follow the mechanism-specific safety or data-validity rule.	Harness/connector ICD review, miswire test, retention test and channel isolation measurement.
AGVSAFE-123	Targeted regression for this topic shall use Harness/connector ICD review, miswire test, retention test and channel isolation measurement whenever connector, harness, grounding, shield termination or pin assignment changes.	Harness/connector ICD review, miswire test, retention test and channel isolation measurement.

Verification, margin and future-change rule

Changes to connector, harness, grounding, shield termination or pin assignment are screened against the failure mechanism described above, with only demonstrably unaffected evidence carried forward.

MANUFACTURING EOL - SAFETY I/O, PROGRAMMING & TRACEABILITY

EOL verifies hardware identity, safety software, parameter package, safety inputs, STO outputs, communication variant, watchdog behavior and required diagnostics before unit release.

Missing or failed EOL evidence blocks the unit; manufacturing does not mark the controller as operationally degraded to bypass release criteria.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-124	EOL verifies hardware identity, safety software, parameter package, safety inputs, STO outputs, communication variant, watchdog behavior and required diagnostics before unit release.	Serialized EOL record, test limits, software/config checksum and genealogy audit.
AGVSAFE-125	Release status for this topic shall remain open until all required objective evidence is complete and approved for the exact configuration.	Serialized EOL record, test limits, software/config checksum and genealogy audit.
AGVSAFE-126	Any evidence reuse following a change to EOL fixture, software image, parameter package, safety I/O circuit or test limit shall be justified against Serialized EOL record, test limits, software/config checksum and genealogy audit.	Serialized EOL record, test limits, software/config checksum and genealogy audit.

Verification, margin and future-change rule

If EOL fixture, software image, parameter package, safety I/O circuit or test limit changes, the release owner documents the affected assumptions, selects the required regression and updates the evidence index before sign-off.

FAULT-INJECTION QUALIFICATION OF SAFETY CHANNELS

Production and qualification fault injection covers representative open/short, stuck, delayed, corrupted and disagreement conditions across safety I/O, sensors, compute, communication and output paths according to the safety plan.

Test coverage is tied to the credited diagnostic mechanisms; unexercised critical mechanisms remain open safety-case items.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-127	Production and qualification fault injection covers representative open/short, stuck, delayed, corrupted and disagreement conditions across safety I/O, sensors, compute, communication and output paths according to the safety plan.	Fault-injection matrix, observed safe-state timing and diagnostic traceability.
AGVSAFE-128	Incomplete, failed or variant-mismatched evidence for this topic shall block release of the affected configuration; it shall not be represented as a fictitious runtime degraded state.	Fault-injection matrix, observed safe-state timing and diagnostic traceability.
AGVSAFE-129	The release package shall contain Fault-injection matrix, observed safe-state timing and diagnostic traceability for the exact hardware/software variant.	Fault-injection matrix, observed safe-state timing and diagnostic traceability.

Verification, margin and future-change rule

Variant evolution involving fault model, diagnostic mechanism, channel implementation or safety architecture requires a traceable change-impact review that identifies what remains valid and what must be reverified.

PRODUCTION GENEALOGY, SAFETY CONFIGURATION & CERTIFICATE IDENTITY

Production records link hardware, safety MCU, software/RTOS, safety parameter set, network variant, sensor compatibility and any certification-relevant component identity.

A variant not covered by the released safety evidence is not considered production equivalent.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-130	Production records link hardware, safety MCU, software/RTOS, safety parameter set, network variant, sensor compatibility and any certification-relevant component identity.	Configuration audit, serialized genealogy and certification-scope cross-check.
AGVSAFE-131	The affected hardware or software variant shall remain unreleased while required qualification or production evidence for this topic is open.	Configuration audit, serialized genealogy and certification-scope cross-check.
AGVSAFE-132	Production release is contingent on completion of Configuration audit, serialized genealogy and certification-scope cross-check.	Configuration audit, serialized genealogy and certification-scope cross-check.

Verification, margin and future-change rule

The verification baseline is reopened when BOM, firmware, RTOS, parameter set, network stack or safety-certified component changes enough to alter the stated behavior, margin, interface or failure response.

DVP - STOPPING DISTANCE, PAYLOAD & FLOOR CONDITIONS

Stopping DVP validates protective-field assumptions over released speed, payload, floor/friction, tire and brake conditions. Measured stopping performance is correlated to the safety model and margin.

A single nominal warehouse-floor test is insufficient to substantiate all released variants.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-133	Stopping DVP validates protective-field assumptions over released speed, payload, floor/friction, tire and brake conditions.	Instrumented stopping tests, worst-case analysis and protective-field verification.
AGVSAFE-134	A missing verification result, manufacturing record or similarity justification for this topic is a release-gate condition and shall not be converted into runtime behavior.	Instrumented stopping tests, worst-case analysis and protective-field verification.
AGVSAFE-135	Engineering approval after changes to payload, floor, tire, brake, speed limit or protective field shall be based on Instrumented stopping tests, worst-case analysis and protective-field verification.	Instrumented stopping tests, worst-case analysis and protective-field verification.

Verification, margin and future-change rule

Any modification of payload, floor, tire, brake, speed limit or protective field is compared with the released assumptions on this page; unproven similarity results in targeted re-analysis or retest.

DVP - STO / SLS FUNCTIONAL & TIMING VERIFICATION

STO and SLS DVP measures fault/request-to-safe-output timing, overspeed detection, limit hysteresis, restart interlocks and drive response under representative power and processing load.

Acceptance is tied to the released FTTI and machine safety concept rather than a generic test duration.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-136	Engineering sign-off for this topic shall confirm: STO and SLS DVP measures fault/request-to-safe-output timing, overspeed detection, limit hysteresis, restart interlocks and drive response under representative power and processing load.	Safety-function DVP report with timing traces and pass/fail limits.
AGVSAFE-137	Release status for this topic shall remain open until all required objective evidence is complete and approved for the exact configuration.	Safety-function DVP report with timing traces and pass/fail limits.
AGVSAFE-138	Release approval for this topic shall require: Safety-function DVP report with timing traces and pass/fail limits.	Safety-function DVP report with timing traces and pass/fail limits.

Verification, margin and future-change rule

Future changes in STO/SLS implementation, drive interface, FTTI, processor load or power condition are accepted only after the configuration record explains evidence reuse, new risks and the regression needed for closure.

DVP - SENSOR LOSS, COMMUNICATION FAULTS & DEGRADED STATES

DVP exercises LiDAR loss/obscuration, encoder faults, safety-network timeout, cross-channel disagreement and approved degraded modes while confirming safe-state priority.

The objective is correct safety containment; navigation mission completion is secondary.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-139	DVP exercises LiDAR loss/obscuration, encoder faults, safety-network timeout, cross-channel disagreement and approved degraded modes while confirming safe-state priority.	Scenario DVP, fault logs, state-transition evidence and recovery verification.
AGVSAFE-140	Incomplete, failed or variant-mismatched evidence for this topic shall block release of the affected configuration; it shall not be represented as a fictitious runtime degraded state.	Scenario DVP, fault logs, state-transition evidence and recovery verification.
AGVSAFE-141	A change to sensor, network profile, fallback logic or degraded-mode allocation shall reopen the affected release evidence; closure shall include Scenario DVP, fault logs, state-transition evidence and recovery verification.	Scenario DVP, fault logs, state-transition evidence and recovery verification.

Verification, margin and future-change rule

Engineering change control for sensor, network profile, fallback logic or degraded-mode allocation preserves the intent of this page by linking the modified design to updated analysis, test evidence and variant identity.

POWER-CYCLE, RESET, STARTUP SELF-TEST & LATENT FAULT DVP

Repeated startup, brownout and reset tests verify that self-tests complete, latent faults are detected as intended, outputs remain safe during initialization and no bypass state survives unexpectedly.

Power cycling is performed across selected voltage and temperature corners because reset behavior can be condition dependent.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-142	No affected variant may be released until this condition is satisfied: Repeated startup, brownout and reset tests verify that self-tests complete, latent faults are detected as intended, outputs remain safe during initialization and no bypass state survives unexpectedly.	Power-cycle matrix, startup trace, retained-fault inspection and output-state capture.
AGVSAFE-143	The affected hardware or software variant shall remain unreleased while required qualification or production evidence for this topic is open.	Power-cycle matrix, startup trace, retained-fault inspection and output-state capture.
AGVSAFE-144	Production equivalence after a change to power architecture, reset supervisor, self-test or retained diagnostics shall require Power-cycle matrix, startup trace, retained-fault inspection and output-state capture.	Power-cycle matrix, startup trace, retained-fault inspection and output-state capture.

Verification, margin and future-change rule

A change to power architecture, reset supervisor, self-test or retained diagnostics shall not inherit prior approval automatically; the affected evidence is reviewed for continued representativeness before release.

ENVIRONMENTAL SEQUENTIAL-STRESS QUALIFICATION

Qualification combines the selected thermal, vibration, shock, EMC and power stresses in a sequence that can expose accumulated damage, followed by complete safety-function retest.

Passing each stress separately does not prove the released assembly remains safe after combined aging.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-145	Release approval shall enforce the following condition: Qualification combines the selected thermal, vibration, shock, EMC and power stresses in a sequence that can expose accumulated damage, followed by complete safety-function retest.	Sequential-stress qualification plan and post-stress safety DVP results.
AGVSAFE-146	A missing verification result, manufacturing record or similarity justification for this topic is a release-gate condition and shall not be converted into runtime behavior.	Sequential-stress qualification plan and post-stress safety DVP results.
AGVSAFE-147	The released configuration index shall reference the following objective evidence: Sequential-stress qualification plan and post-stress safety DVP results.	Sequential-stress qualification plan and post-stress safety DVP results.

Verification, margin and future-change rule

Changes affecting environmental profile, enclosure, connector, PCB or mounting stack require documented impact analysis against the mechanism on this page; reusable evidence is retained only within its demonstrated similarity boundary.

FUNCTIONAL SAFETY CASE, SIL 3 / PL e EVIDENCE & LIMITATIONS

The product carries a SIL 3 / PL e STO and SLS release target. This PRD requires the responsible program to provide the applicable certification evidence, scope, assumptions of use and residual limitations before any certified status is claimed; candidate standards remain subject to program applicability.

Component certificates or prior-use evidence are reused only when configuration and assumptions match the released AGV application.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-148	The product carries a SIL 3 / PL e STO and SLS release target. This PRD requires the responsible program to provide the applicable certification evidence, scope, assumptions of use and residual limitations before any certified status is claimed; candidate standards remain subject to program applicability.	Safety case, certification reports/certificates, assumption-of-use closure and independent review.
AGVSAFE-149	Release status for this topic shall remain open until all required objective evidence is complete and approved for the exact configuration.	Safety case, certification reports/certificates, assumption-of-use closure and independent review.
AGVSAFE-150	This topic remains open in the release matrix until Safety case, certification reports/certificates, assumption-of-use closure and independent review is approved.	Safety case, certification reports/certificates, assumption-of-use closure and independent review.

Verification, margin and future-change rule

Reopen this verification topic whenever safety standard, certification body/scope, component, architecture or assumption of use changes. The responsible engineer identifies moved assumptions and closes the resulting analysis or DVP before approval.

MACHINERY / AGV STANDARD MAPPING & COMPLIANCE MATRIX

The program maps applicable machinery and driver safety standards such as ISO 3691-4, IEC 61800-5-2 and safety-communication standards to product requirements where contractually required. Editions and applicability are TBC until program confirmation.

No standard is silently claimed solely because the architecture uses STO/SLS or a named safety network.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-151	The program maps applicable machinery and driver safety standards such as ISO 3691-4, IEC 61800-5-2 and safety-communication standards to product requirements where contractually required.	Approved standards applicability matrix and compliance evidence map.
AGVSAFE-152	Incomplete, failed or variant-mismatched evidence for this topic shall block release of the affected configuration; it shall not be represented as a fictitious runtime degraded state.	Approved standards applicability matrix and compliance evidence map.
AGVSAFE-153	Release evidence after modifications to market, AGV type, safety function, network or regulatory scope shall include Approved standards applicability matrix and compliance evidence map.	Approved standards applicability matrix and compliance evidence map.

Verification, margin and future-change rule

The change-control trigger here is any modification to market, AGV type, safety function, network or regulatory scope. Acceptance requires restored margin, interface consistency and objective evidence for the affected mechanism.

15-YEAR LIFECYCLE, SAFETY-STOP ANALYTICS & FIELD FEEDBACK

The baseline assigns a 15-year lifecycle with continuous safety-stop logging. Field data supports hazard analytics, maintenance and product improvement while safety parameter changes remain controlled engineering releases.

Trend analytics do not automatically modify safety limits or protective fields in service. Any such change follows the safety-change process and revalidation.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-154	The baseline assigns a 15-year lifecycle with continuous safety-stop logging.	Lifecycle plan, field-event governance, parameter-change records and obsolescence register.
AGVSAFE-155	The affected hardware or software variant shall remain unreleased while required qualification or production evidence for this topic is open.	Lifecycle plan, field-event governance, parameter-change records and obsolescence register.
AGVSAFE-156	The program shall close Lifecycle plan, field-event governance, parameter-change records and obsolescence register before approving the affected production variant.	Lifecycle plan, field-event governance, parameter-change records and obsolescence register.

Verification, margin and future-change rule

For future variants, a change in field analytics, safety parameter, logging backend, component life or maintenance policy is an engineering event, not an editorial update; the impact record defines evidence reuse and targeted regression.

FIELD DIAGNOSTICS, SERVICE REPLACEMENT & RE-COMMISSIONING

Replacement of controller, LiDAR, encoder, drive interface or safety network component requires the commissioning checks necessary to restore configuration identity and safety assumptions.

A service replacement is not presumed equivalent merely because connectors mate and communication starts.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-157	Replacement of controller, LiDAR, encoder, drive interface or safety network component requires the commissioning checks necessary to restore configuration identity and safety assumptions.	Service procedure, replacement simulation, parameter restore and functional safety check.
AGVSAFE-158	A missing verification result, manufacturing record or similarity justification for this topic is a release-gate condition and shall not be converted into runtime behavior.	Service procedure, replacement simulation, parameter restore and functional safety check.
AGVSAFE-159	Variant sign-off shall reference Service procedure, replacement simulation, parameter restore and functional safety check and the applicable configuration identity.	Service procedure, replacement simulation, parameter restore and functional safety check.

Verification, margin and future-change rule

Changes to field-replaceable unit, sensor, drive, network component or commissioning tool are screened against the failure mechanism described above, with only demonstrably unaffected evidence carried forward.

SUPPLIER CHANGE, SECOND SOURCE & SAFETY IMPACT ASSESSMENT

Safety-relevant supplier changes are screened for electrical, timing, diagnostic, environmental and systematic effects. Evidence reuse is justified explicitly, including certification scope where relevant.

Production equivalence remains blocked until affected safety analysis and regression DVP are closed.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-160	Safety-relevant supplier changes are screened for electrical, timing, diagnostic, environmental and systematic effects.	Supplier impact assessment, similarity justification, updated safety case and regression test.
AGVSAFE-161	Release status for this topic shall remain open until all required objective evidence is complete and approved for the exact configuration.	Supplier impact assessment, similarity justification, updated safety case and regression test.
AGVSAFE-162	Any evidence reuse following a change to MCU, sensor, output driver, network ASIC, power device, connector or supplier process shall be justified against Supplier impact assessment, similarity justification, updated safety case and regression test.	Supplier impact assessment, similarity justification, updated safety case and regression test.

Verification, margin and future-change rule

If MCU, sensor, output driver, network ASIC, power device, connector or supplier process changes, the release owner documents the affected assumptions, selects the required regression and updates the evidence index before sign-off.

FINAL RELEASE CLOSURE, OPEN-POINT DISPOSITION & SAFETY SIGN-OFF

Release requires closed safety requirements traceability, timing evidence, DVP, certification scope, manufacturing controls, configuration genealogy, cybersecurity boundaries, lifecycle ownership and approved deviations.

Open qualification or safety-case gaps remain visible release blockers and are not disguised as operational degraded states.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
AGVSAFE-163	Release requires closed safety requirements traceability, timing evidence, DVP, certification scope, manufacturing controls, configuration genealogy, cybersecurity boundaries, lifecycle ownership and approved deviations.	Final release checklist, signed safety review, deviation log and configuration index.
AGVSAFE-164	Incomplete, failed or variant-mismatched evidence for this topic shall block release of the affected configuration; it shall not be represented as a fictitious runtime degraded state.	Final release checklist, signed safety review, deviation log and configuration index.
AGVSAFE-165	The release package shall contain Final release checklist, signed safety review, deviation log and configuration index for the exact hardware/software variant.	Final release checklist, signed safety review, deviation log and configuration index.

Verification, margin and future-change rule

Variant evolution involving any released safety baseline element or unresolved evidence item requires a traceable change-impact review that identifies what remains valid and what must be reverified.